Networking Module 5

S Vimal

1) Capture and analyze ARP packets using Wireshark. Inspect the ARP request and reply frames, and discuss the role of the sender's IP and MAC address in these packets.

ARP (Address Resolution Protocol) operates at the boundary of Layer 2 and Layer 3 of the OSI model. It resolves a known IP address into its corresponding MAC address within a local network. Since data link layer communication relies on MAC addresses to deliver frames, ARP acts as the essential bridge between logical IP addressing and physical hardware addressing. Without ARP, devices on the same subnet would be unable to communicate even if they knew each other's IP addresses.

When a device needs to communicate with another on the same network, it first checks its ARP cache. If no entry is found, it broadcasts an ARP request to all devices using destination MAC FF:FF:FF:FF:FF:FF. Only the device owning the requested IP responds with a unicast ARP reply containing its MAC address. The requesting device then stores this mapping in its ARP cache for future use.

The sender's IP and MAC address play a critical role here. The sender's IP identifies the logical network identity of the requesting device, while the sender's MAC identifies its physical hardware interface. The target device uses the sender's MAC to send the reply directly back, and both devices update their ARP tables with this information for faster future communication.

Sender MAC Address: Physical address of the requesting device. The target uses this to send the reply directly back via unicast.

Sender IP Address: Logical address of the requesting device. The target reads this to identify who is asking and updates its own ARP cache accordingly.

Target MAC Address: Set to 00:00:00:00:00:00 in a request, since this is the unknown value being discovered.

Target IP Address: The IP address being queried. Only the device matching this IP will respond.

Packet from a sender to an IP 10.0.2.2

No.	Time	Source	Destination	Protocol	Length	Info
19	5.101569020	PCSSystemtec_f5:68:…	52:54:00:12:35:02	ARP	42	Who h
20	5.102050261	52:54:00:12:35:02	PCSSystemtec_f5:68:…	ARP	60	10.0.

```
▸ Frame 19: 42 bytes on wire (336 bits), 42 bytes captured (336 bits) on interfac
▸ Ethernet II, Src: PCSSystemtec_f5:68:17 (08:00:27:f5:68:17), Dst: 52:54:00:12:3
▾ Address Resolution Protocol (request)
    Hardware type: Ethernet (1)
    Protocol type: IPv4 (0x0800)
    Hardware size: 6
    Protocol size: 4
    Opcode: request (1)
    Sender MAC address: PCSSystemtec_f5:68:17 (08:00:27:f5:68:17)
    Sender IP address: 10.0.2.15
    Target MAC address: 00:00:00_00:00:00 (00:00:00:00:00:00)
    Target IP address: 10.0.2.2
```

Reply from IP 10.0.2.2 with its mac address

19	5.101569020	PCSSystemtec_f5:68:…	52:54:00:12:35:02	ARP	42	Who
20	5.102050261	52:54:00:12:35:02	PCSSystemtec_f5:68:…	ARP	60	10.

```
▸ Frame 20: 60 bytes on wire (480 bits), 60 bytes captured (480 bits) on interfac
▸ Ethernet II, Src: 52:54:00:12:35:02 (52:54:00:12:35:02), Dst: PCSSystemtec_f5:6
▾ Address Resolution Protocol (reply)
    Hardware type: Ethernet (1)
    Protocol type: IPv4 (0x0800)
    Hardware size: 6
    Protocol size: 4
    Opcode: reply (2)
    Sender MAC address: 52:54:00:12:35:02 (52:54:00:12:35:02)
    Sender IP address: 10.0.2.2
    Target MAC address: PCSSystemtec_f5:68:17 (08:00:27:f5:68:17)
    Target IP address: 10.0.2.15
```

2. Using Packet Tracer, simulate an ARP spoofing attack. Analyse the behaviour of devices on the network when they receive a malicious ARP response.

A spoofing attack can be demonstrated using , PC0 as a victim, PC1 as a gateway and PC2 as an attacker.

PC0:

IP: 10.10.10.1

Subnet: 255.0.0.0

PC1:

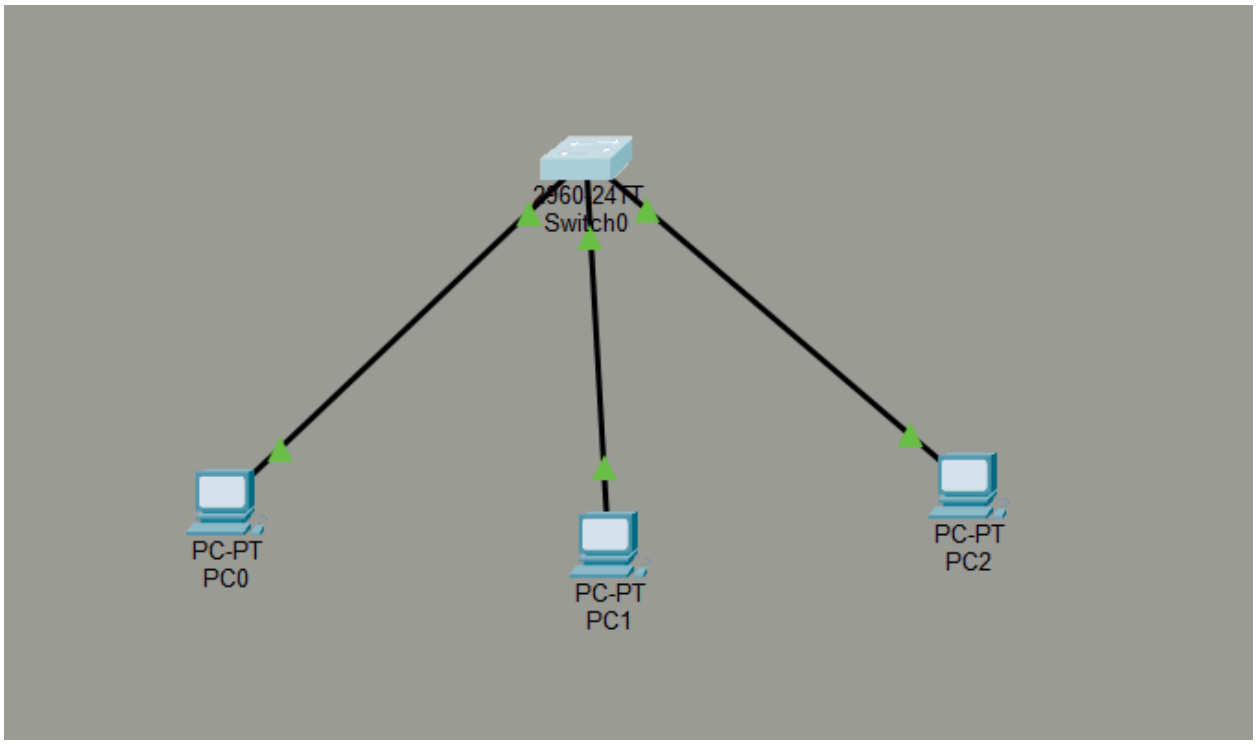
IP:10.10.10.2

Subnet: 255.0.0.0

PC2:

IP:10.10.10.2

Subnet: 255.0.0.0



Normal Behaviour without spoofing, pinging from PC0 to PC1 and checking arp table

PC0

Physical | Config | Desktop | Programming | Attributes

Command Prompt

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 10.10.10.2

Pinging 10.10.10.2 with 32 bytes of data:

Reply from 10.10.10.2: bytes=32 time=1ms TTL=128
Reply from 10.10.10.2: bytes=32 time<1ms TTL=128
Reply from 10.10.10.2: bytes=32 time<1ms TTL=128
Reply from 10.10.10.2: bytes=32 time<1ms TTL=128

Ping statistics for 10.10.10.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>arp -a
  Internet Address      Physical Address      Type
  10.10.10.2            00d0.ff08.505c        dynamic
```

<u>Attacker sends a fake ARP reply</u>

After pinging the PC1 from PC0

PDU Information at Device: PC2

OSI Model | Inbound PDU Details | Outbound PDU Details

At Device: PC2
Source: PC0
Destination: Broadcast

In Layers	Out Layers
Layer7	Layer7
Layer6	Layer6
Layer5	Layer5
Layer4	Layer4
Layer3	Layer3
Layer 2: Ethernet II Header 0005.5E2E.B2E5 >> FFFF.FFFF.FFFF ARP Packet Src. IP: 10.10.10.1, Dest. IP: 10.10.10.2	Layer 2: Ethernet II Header 0050.0FD7.2D6D >> 0005.5E2E.B2E5 ARP Packet Src. IP: 10.10.10.2, Dest. IP: 10.10.10.1
Layer 1: Port FastEthernet0	Layer 1: Port(s): FastEthernet0

1. FastEthernet0 receives the frame.

After a spoofing attack, the packet meant for PC1 is received by PC2, by this way data can stealed.

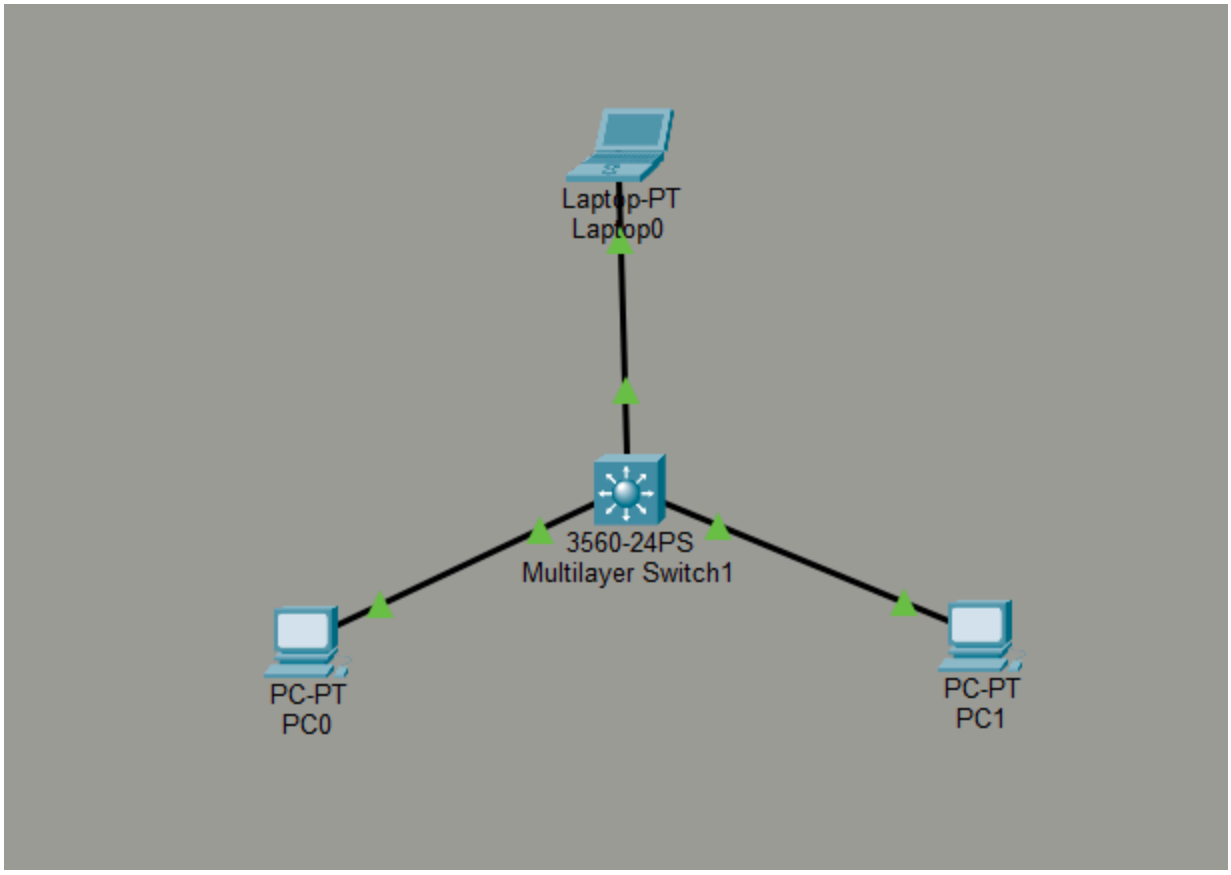
The ARP process was observed in Cisco Packet Tracer using the Simulation Mode with the ARP protocol filter enabled. A small network was set up with three PCs connected through a switch, each assigned an IP address within the same subnet. When a ping command was executed from one PC to another, the ARP request and reply packets were captured and analyzed using the OSI Model view.

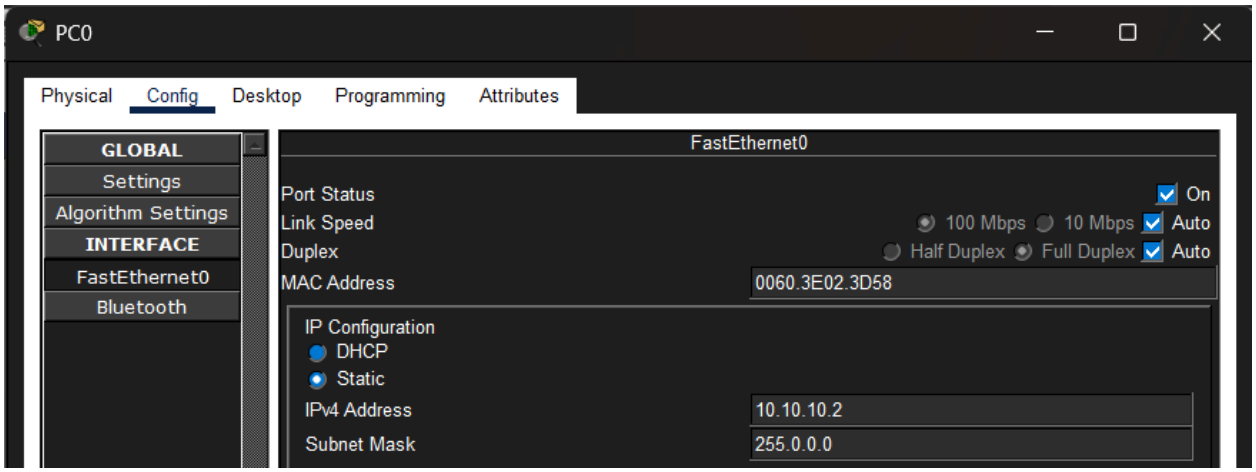
The ARP request was broadcast from PC2 with the destination MAC address FFFF.FFFF.FFFF, meaning the frame was sent to all devices on the network. The switch received the frame and forwarded it to all connected ports except the one it arrived on. The device matching the requested IP address then responded with an ARP reply containing its MAC address, allowing the requesting device to update its ARP table and proceed with communication.

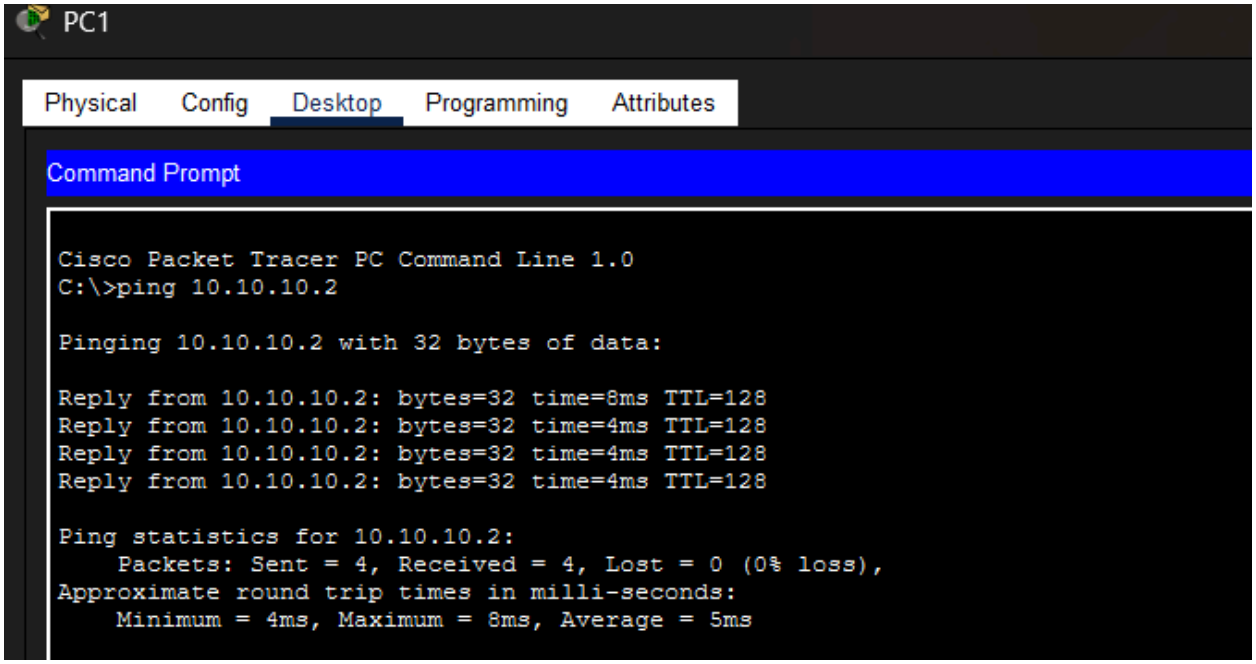
ARP has no authentication mechanism, which makes it vulnerable to spoofing attacks. In an ARP spoofing attack, a malicious device sends a forged ARP reply claiming that its MAC address belongs to another device's IP address, such as the default gateway. Since ARP replies are accepted without verification, the victim device updates its ARP table with the incorrect mapping. This causes traffic intended for the gateway to be redirected through the attacker's machine instead, enabling interception or manipulation of network data. This type of attack is also known as a Man-in-the-Middle attack and is one of the most common Layer 2 security threats in local networks.

3. Manually configure static IPs on the client device (like PC or your mobile phone) and verify connectivity using ping.

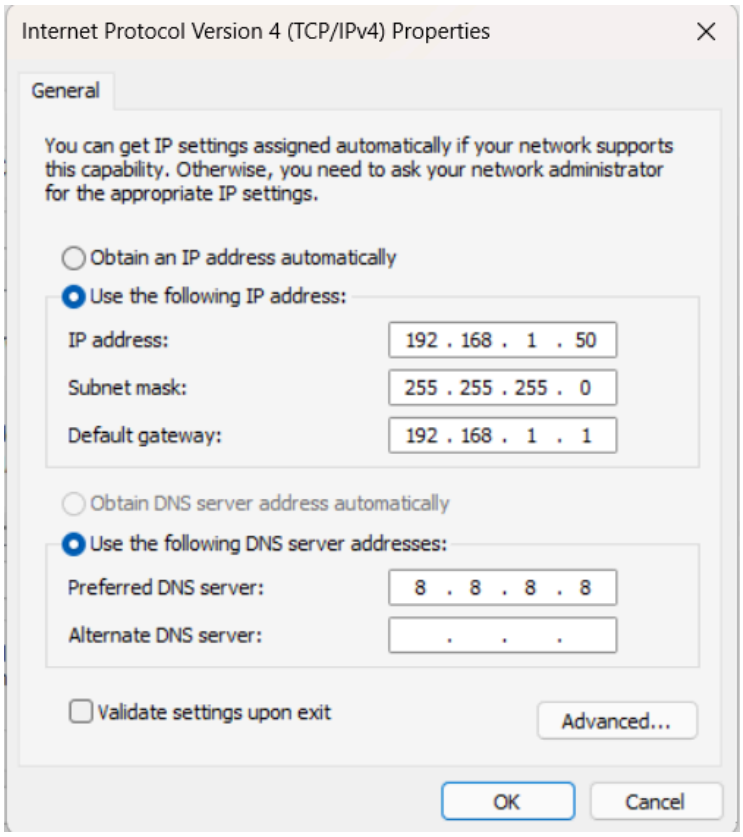
In cisco packet tracer, IP addresses of client devices are manually configured (static) and then pinged to test connectivity.


Laptop-PT
Laptop0
3560-24PS
Multilayer Switch1
PC-PT
PC0
PC-PT
PC1


PC0
Physical
Config
Desktop
Programming
Attributes
GLOBAL
Settings
Algorithm Settings
INTERFACE
FastEthernet0
Bluetooth
FastEthernet0
Port Status
On
Link Speed
100 Mbps
10 Mbps
Auto
Duplex
Half Duplex
Full Duplex
Auto
MAC Address
0060.3E02.3D58
IP Configuration
DHCP
Static
IPv4 Address
10.10.10.2
Subnet Mask
255.0.0.0



In Windows Laptop, assigning a dummy IP address to the PC, manually



Checking this using ping

```
C:\Users\vimal>ping 8.8.8.8

Pinging 8.8.8.8 with 32 bytes of data:
Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 8.8.8.8:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\Users\vimal>ping 142.250.207.68

Pinging 142.250.207.68 with 32 bytes of data:
Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 142.250.207.68:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
```

It cannot ping these IP addresses.

4. Use Wireshark to capture DHCP Discover, Offer, Request, and Acknowledge messages and explain the process.

DHCP (Dynamic Host Configuration Protocol) automates the assignment of IP addresses to devices on a network. When a device connects to a network, it does not have an IP address yet and needs to request one from a DHCP server. This entire negotiation happens through a four-step process known as DORA, Discover, Offer, Request, and Acknowledge. The process completes within seconds and results in the client receiving a valid IP address, subnet mask, default gateway, and DNS server information automatically.

Using the command sudo dhclient enp0s3 in ubuntu DORA process is started and wireshark captures the packet.

526	157.202222071	0.0.0.0	255.255.255.255	DHCP	342	DHCP Discover - Transaction ID 0x771fed6c
527	157.202776826	10.0.2.2	10.0.2.15	DHCP	590	DHCP Offer - Transaction ID 0x771fed6c
528	157.202904719	0.0.0.0	255.255.255.255	DHCP	342	DHCP Request - Transaction ID 0x771fed6c
529	157.203099273	10.0.2.2	10.0.2.15	DHCP	590	DHCP ACK - Transaction ID 0x771fed6c

```
▸ Frame 526: 342 bytes on wire (2736 bits), 342 bytes captured (2736 bits) on interface enp0s3, id 0
▸ Ethernet II, Src: PCSSystemtec_f5:68:17 (08:00:27:f5:68:17), Dst: Broadcast (ff:ff:ff:ff:ff:ff)
▸ Internet Protocol Version 4, Src: 0.0.0.0, Dst: 255.255.255.255
▸ User Datagram Protocol, Src Port: 68, Dst Port: 67
▸ Dynamic Host Configuration Protocol (Discover)
```

DHCP Discover: A client without an IP address broadcasts a message across the network to locate an available DHCP server. (broadcast)
DHCP Offer: A DHCP server responds to the broadcast by proposing an available IP address and network configuration details to the client. (unicast)
DHCP Request: The client formally accepts the offered IP address and broadcasts this decision so all servers on the network know the offer was claimed. (broadcast)
DHCP Acknowledge: The server confirms the assignment and finalizes the configuration, completing the process and allowing the client to communicate. (unicast)

5. Given an IP address range of 192.168.1.0/24, divide the network into 4 subnets.

Task:

- **Manually calculate the new subnet mask and the range of valid IP addresses for each subnet.**
- **Assign IP addresses from these subnets to devices in Cisco Packet Tracer and verify connectivity using ping between them.**

<u>Manual Calculation</u>

(5) The Given,

IP : 192.168.1.0 / 24.

∴ Need to be divided into 4 subnets.

→ Total addresses possible $= 2^{32-24} = 2^8 = 256$ addresses.

$= \frac{256}{4} = 64 \Rightarrow 2^6$

The CIDR is /26 i.e $(32-26=6)$.

∴ The new subnet mask is 255.255.255.192

Subnet 1:

Network address = 192.168.1.0
Broadcast address = 192.168.1.63
First valid address = 192.168.1.1
Last ~~Second~~ valid address = 192.168.1.62.

Subnet 2:

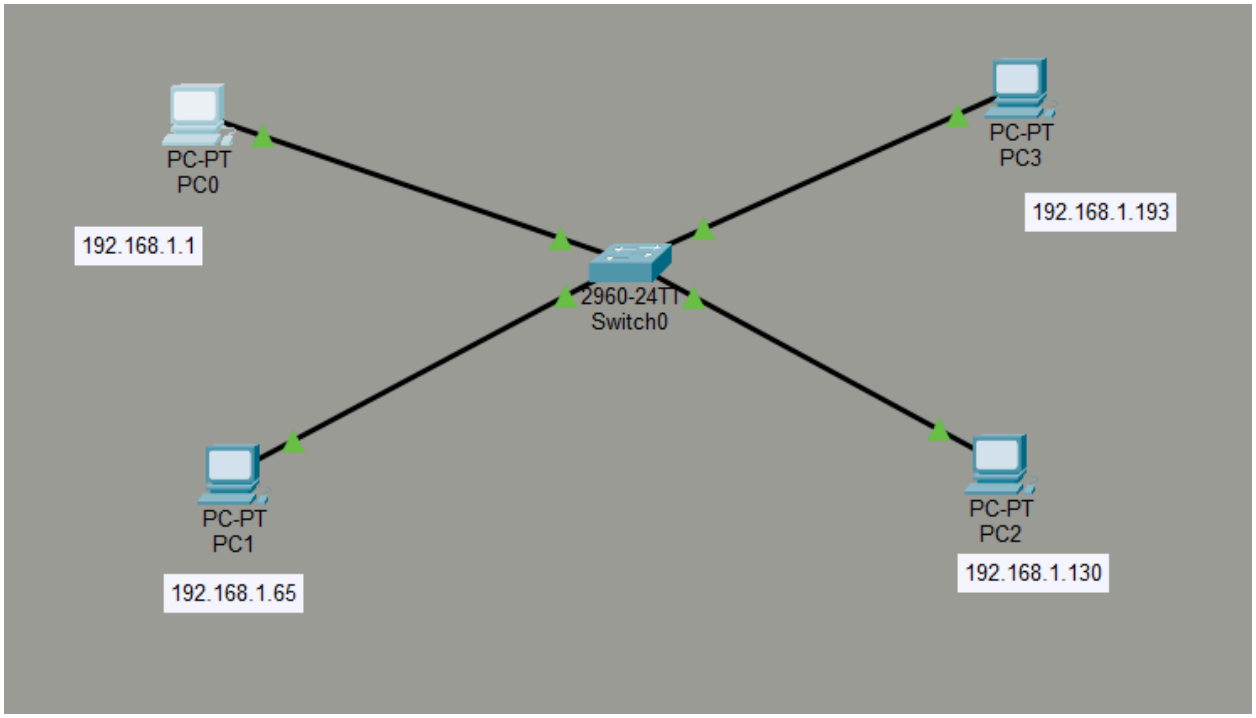
Network address = 192.168.1.64
Broadcast address = 192.168.1.127
First valid address = 192.168.1.65
Last valid address : 192.168.1.126

Subnet-3

Network address :- 192.168.1.128
Broadcast address :- 192.168.1.191
First valid address - 192.168.1.129
Last valid address. - 192.168.1.190

Subnet-4

Network Address - 192.168.1.192.
Broadcast Address - 192.168.1.255
First valid Address - 192.168.1.193
Last valid Address - 192.168.1.254.



Pinging

PC0 TO PC1

```
C:\>ping 192.168.1.65

Pinging 192.168.1.65 with 32 bytes of data:

Reply from 192.168.1.65: bytes=32 time<1ms TTL=128
Reply from 192.168.1.65: bytes=32 time<1ms TTL=128
Reply from 192.168.1.65: bytes=32 time<1ms TTL=128
Reply from 192.168.1.65: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.1.65:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

PC0 TO PC2

```
C:\>ping 192.168.1.130

Pinging 192.168.1.130 with 32 bytes of data:

Reply from 192.168.1.130: bytes=32 time<1ms TTL=128
Reply from 192.168.1.130: bytes=32 time<1ms TTL=128
Reply from 192.168.1.130: bytes=32 time<1ms TTL=128
Reply from 192.168.1.130: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.1.130:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

PC0 TO PC3

```
C:\>ping 192.168.1.193

Pinging 192.168.1.193 with 32 bytes of data:

Reply from 192.168.1.193: bytes=32 time<1ms TTL=128
Reply from 192.168.1.193: bytes=32 time<1ms TTL=128
Reply from 192.168.1.193: bytes=32 time=3ms TTL=128
Reply from 192.168.1.193: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.1.193:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 3ms, Average = 0ms
```

PC2 TO PC3

PC2

Physical | Config | Desktop | Programming | Attributes

Command Prompt

```
^C
C:\>ping 192.168.1.193

Pinging 192.168.1.193 with 32 bytes of data:

Reply from 192.168.1.193: bytes=32 time<1ms TTL=128
Reply from 192.168.1.193: bytes=32 time=3ms TTL=128
Reply from 192.168.1.193: bytes=32 time<1ms TTL=128
Reply from 192.168.1.193: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.1.193:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 3ms, Average = 0ms
```

PC2 TO PC1

```
C:\>ping 192.168.1.65

Pinging 192.168.1.65 with 32 bytes of data:

Reply from 192.168.1.65: bytes=32 time<1ms TTL=128
Reply from 192.168.1.65: bytes=32 time<1ms TTL=128
Reply from 192.168.1.65: bytes=32 time<1ms TTL=128
Reply from 192.168.1.65: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.1.65:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

PC1 TO PC3

PC1

Physical | Config | Desktop | Programming | Attributes

Command Prompt

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.1.193

Pinging 192.168.1.193 with 32 bytes of data:

Reply from 192.168.1.193: bytes=32 time<1ms TTL=128
Reply from 192.168.1.193: bytes=32 time<1ms TTL=128
Reply from 192.168.1.193: bytes=32 time<1ms TTL=128
Reply from 192.168.1.193: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.1.193:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

Four subnets were created from the network 192.168.1.0/24. For verification, devices were assigned addresses from the same subnet to test connectivity using the ping command.

6. You are given three IP addresses: 10.1.1.1, 172.16.5.10, and 192.168.1.5.

Task:

- **Identify the class of each IP address (Class A, B, or C).**
- **What is the default subnet mask for each class?**
- **Provide the range of IP addresses for each class.**

IP Address	Class	Default Subnet	IP address range
10.1.1.1	**Class A**	**255.0.0.0**	**1.0.0.0 – 126.255.255.255**
172.16.5.10	**Class B**	**255.0.0.0**	**128.0.0.0 – 191.255.255.255**
192.168.1.5	**Class C**	**255.255.255.0**	**192.0.0.0 – 223.255.255.255**

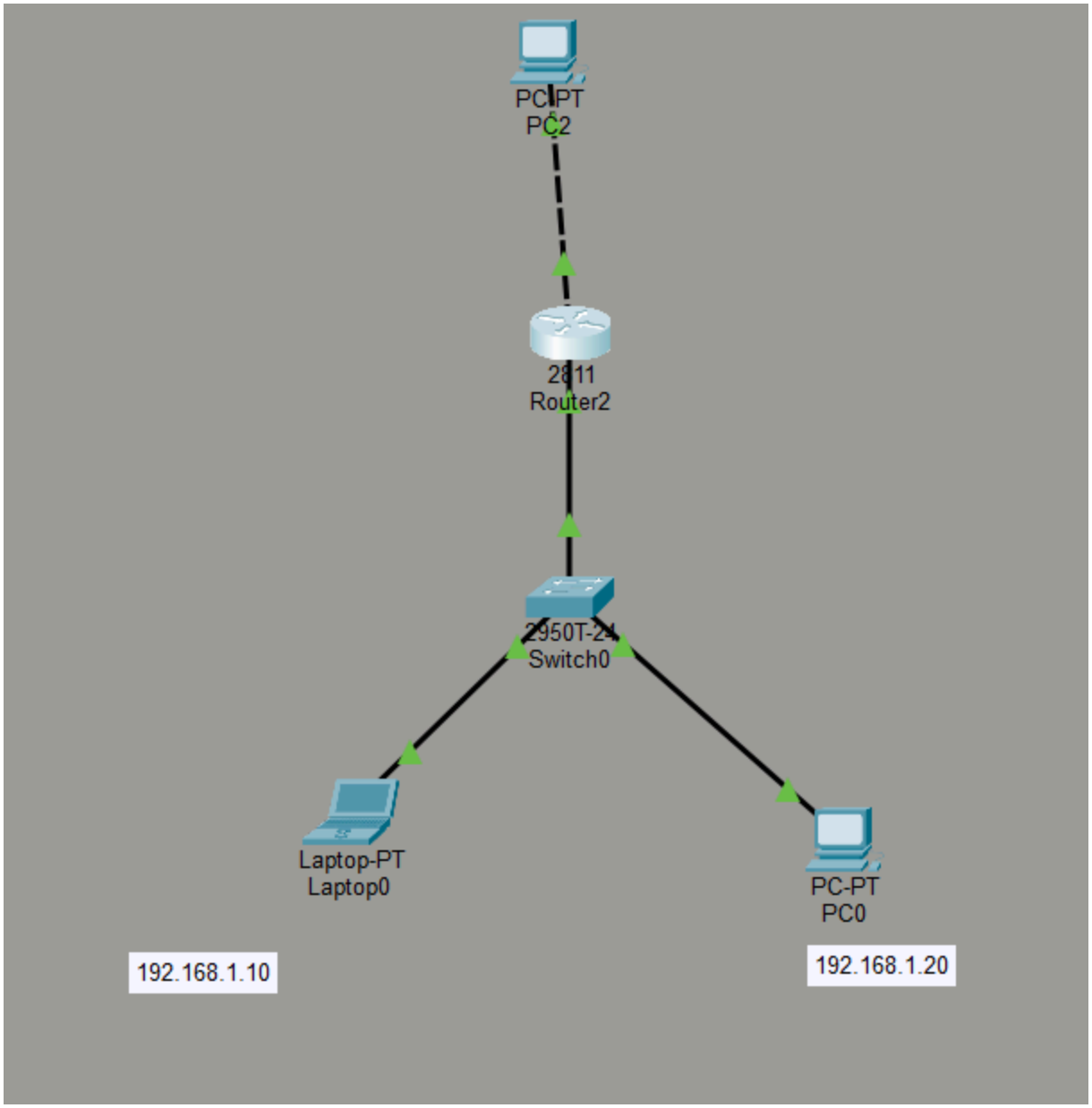
The classification of an IP address depends on the value of its first octet. Taking the address 10.1.1.1 as an example, its first octet is 10, which falls between 1 and 126, placing it under Class A with a default subnet mask of 255.0.0.0. For the address 172.16.5.10, the first octet 172 sits in the range of 128 to 191, which identifies it as a Class B address carrying a default subnet mask of 255.255.0.0. Finally, 192.168.1.5 has a first octet of 192, which lies between 192 and 223, making it a Class C address with a default subnet mask of 255.255.255.0.

7) In Cisco Packet Tracer, create a small network with multiple devices (e.g., 2 PCs and a router. Use private IP addresses (e.g., 192.168.1 .x) on the PCs and configure the router to perform NAT to allow the PCs to access the internet.

Task:

- **Test the NAT configuration by pinging an external IP address from the PCs and capture the traffic using Wireshark.**
- **What is the source IP address before and after NAT?**

CISCO Setup



LAN Configurations

```
Router(config)#interface FastEthernet0/0
Router(config-if)#ip address 192.168.1.1 255.255.255.0
Router(config-if)#ip nat inside
Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up
exit
```

WAN configurations

```
Router(config)#interface FastEthernet0/1
Router(config-if)#ip address 203.0.113.1 255.255.255.0
Router(config-if)#ip nat outside
Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up
exit
```

NAT configurations

```
Router(config)#access-list 1 permit 192.168.1.0 0.0.0.255
Router(config)#ip nat inside source list 1 interface FastEthernet0/1 overload
```

Pinging external IP

203.0.113.2 is an external IP

```
C:\>ping 203.0.113.2

Pinging 203.0.113.2 with 32 bytes of data:

Request timed out.
Reply from 203.0.113.2: bytes=32 time<1ms TTL=127
Reply from 203.0.113.2: bytes=32 time<1ms TTL=127
Reply from 203.0.113.2: bytes=32 time<1ms TTL=127

Ping statistics for 203.0.113.2:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

NAT Translations

```
Router#show ip nat translations
Pro  Inside global     Inside local       Outside local      Outside global
icmp 203.0.113.1:13    192.168.1.10:13    203.0.113.2:13     203.0.113.2:13
icmp 203.0.113.1:14    192.168.1.10:14    203.0.113.2:14     203.0.113.2:14
icmp 203.0.113.1:15    192.168.1.10:15    203.0.113.2:15     203.0.113.2:15
icmp 203.0.113.1:16    192.168.1.10:16    203.0.113.2:16     203.0.113.2:16
```

Source IP address before and after NAT

IP of source before NAT is: 192.168.1.10

IP of source after NAT is: 203.0.113.1

LAN Side (Inside-FastEthernet0/0): The LAN consists of Laptop0 and PC0 connected through a 2960 switch to the router's inside interface. Both PCs are assigned private IP addresses from

the 192.168.1.0/24 range, which are not routable on the internet. The router's inside interface FastEthernet0/0 is configured with IP 192.168.1.1 and acts as the default gateway for all internal devices. The ip nat inside command is applied on this interface to tell the router that traffic coming from this side needs to be translated before going out.

WAN Side (Outside-FastEthernet0/1): The WAN side is simulated using PC2 acting as an external internet server with IP 203.0.113.2. The router's outside interface FastEthernet0/1 is assigned the public IP 203.0.113.1 with the ip nat outside command applied, marking it as the exit point for translated traffic. When internal PCs ping 203.0.113.2, the router replaces their private source IPs with 203.0.113.1 before forwarding the packet out through this interface, making it appear as if all internal traffic is originating from a single public IP address.